

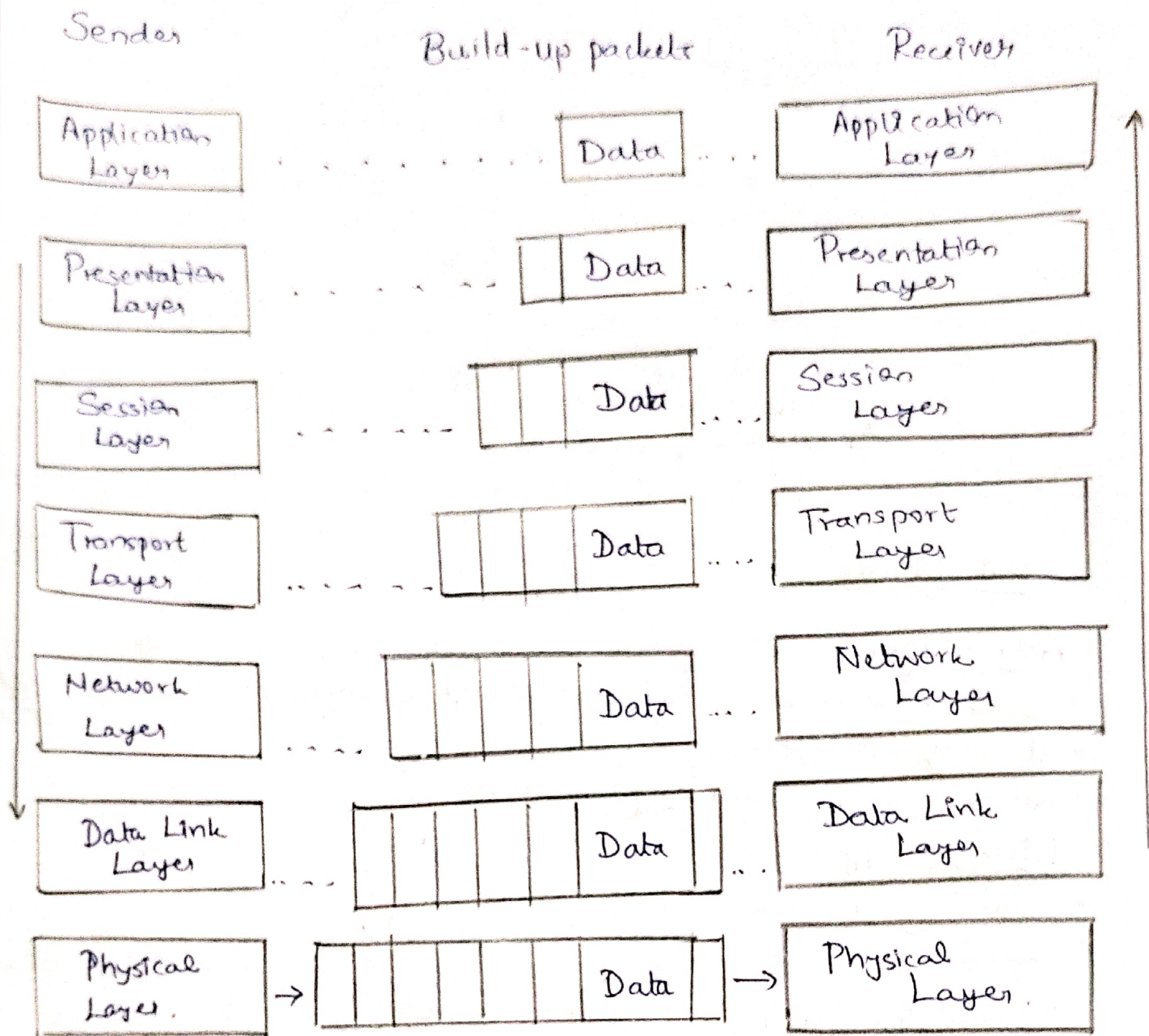
COMPUTER NETWORKS

ASSIGNMENT - 1

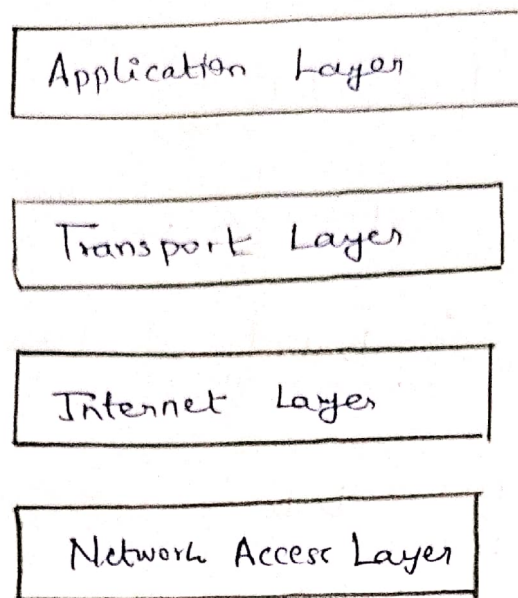
MOHAMED B SIRAJUDEEN

22TD0053

OSI Model



TCP/ IP Model



OSI Model

Advantages

- Both connection-oriented services and connectionless services are supported.
- It is quite flexible
- All the layers work independently,

Disadvantages

- Setting up a model is a challenging task.
- Sometimes, it becomes difficult to fit a new protocol into this model
- It is only used as a reference model.

TCP/IP Model

Advantages

- Many Routing protocols are supported
- It is highly scalable and uses a client-server architecture
- It is lightweight

Disadvantages

- Little difficult to set up.
- Delivery of packets is not guaranteed by the transport layer.
- Vulnerable to a synchronization attack.

Difference between OSI model and TCP/IP Model

Params	OSI Model	TCP/IP Model
Full form	OSI stands for Open Systems Interconnection	TCP/IP stands for Transmission Control Protocol / Internet Protocol
Layers	It has 7 layers	It has 4 layers
Usage	It is low in usage	It is mostly used.
Approach	It is vertically approached	It is horizontally approached
Delivery	Delivery of the package is guaranteed in OSI Model	Delivery of the package is not guaranteed in TCP/IP Model.
Replacement	Replacement of tools and changes can easily be done in this model.	Replacing the tools is not easy as it is in OSI Model.
Reliability	It is less reliable than TCP/IP model	It is more reliable than OSI Model